

1998 年数学(三) 真题解析

一、填空题

(1) 【答案】 e^{-1} .

【解】 $y'|_{x=1} = n$, 曲线 $y = x^n$ 在点 $(1, 1)$ 处的切线为 $y - 1 = n(x - 1)$.

令 $y = 0$ 得 $\xi_n = 1 - \frac{1}{n}$,

则 $\lim_{n \rightarrow \infty} \xi_n^n = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^n = \lim_{n \rightarrow \infty} \left[\left(1 - \frac{1}{n}\right)^{-n}\right]^{-1} = e^{-1}$.

(2) 【答案】 $-\frac{\ln x}{x} + C$.

【解】 $\int \frac{\ln x - 1}{x^2} dx = -\int (\ln x - 1) d\left(\frac{1}{x}\right) = -\frac{\ln x - 1}{x} + \int \frac{1}{x^2} dx = -\frac{\ln x}{x} + C$.

(3) 【答案】 $y_t = C(-5)^t + \frac{5}{12}\left(t - \frac{1}{6}\right)$ (C 为任意常数).

【解】 将 $2y_{t+1} + 10y_t - 5t = 0$ 化为 $y_{t+1} + 5y_t = \frac{5}{2}t$,

$y_{t+1} + 5y_t = 0$ 的通解为 $y_t = C(-5)^t$;

$y_{t+1} + 5y_t = \frac{5}{2}t$ 的特解为 $y^* = at + b$, 代入得 $a = \frac{5}{12}, b = -\frac{5}{72}$,

故 $2y_{t+1} + 10y_t - 5t = 0$ 的通解为 $y_t = C(-5)^t + \frac{5}{12}\left(t - \frac{1}{6}\right)$ (C 为任意常数).

(4) 【答案】 $\begin{pmatrix} 2 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & 2 \end{pmatrix}$.

【解】 $|A| = -2$,

由 $A^*BA = 2BA - 8E$, 得 $AA^*BA = 2ABA - 8A$, 即 $-2BA = 2ABA - 8A$.

整理得

$$BA = -ABA + 4A,$$

因为 A 可逆, 所以由 $BA = -ABA + 4A$, 得 $B = -AB + 4E$,

解得

$$(E + A)B = 4E,$$

故

$$B = 4(E + A)^{-1} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

(5) 【答案】 $\frac{1}{20}, \frac{1}{100}, 2$.

【解】 因为 X_1, X_2, X_3, X_4 服从正态分布 $N(0, 2^2)$ 且相互独立,

所以 $X_1 - 2X_2 \sim N(0, 20)$, $3X_3 - 4X_4 \sim N(0, 100)$,

从而 $\frac{X_1 - 2X_2}{\sqrt{20}} \sim N(0, 1)$, $\frac{3X_3 - 4X_4}{10} \sim N(0, 1)$ 且相互独立,

于是 $\frac{1}{20}(X_1 - 2X_2)^2 + \frac{1}{100}(3X_3 - 4X_4)^2 \sim \chi^2(2)$,

故 $a = \frac{1}{20}, b = \frac{1}{100}, n = 2$.

二、选择题

(1) 【答案】 (D).

【解】 由 $\lim_{x \rightarrow 0} \frac{f(1) - f(1-x)}{2x} = \frac{1}{2} \lim_{x \rightarrow 0} \frac{f(1-x) - f(1)}{-x} = \frac{1}{2} f'(1) = -1$ 得 $f'(1) = -2$,

再由 $f'(5) = f'(1)$ 得 $f'(5) = -2$, 应选(D).

(2) 【答案】 (B).

【解】 当 $|x| < 1$ 时, $f(x) = 1+x$;

当 $|x| > 1$ 时, $f(x) = 0$,

$$\text{再由 } f(-1) = 0, f(1) = 1 \text{ 得 } f(x) = \begin{cases} 1+x, & |x| < 1, \\ 0, & |x| > 1, \\ 0, & x = -1, \\ 1, & x = 1. \end{cases}$$

由 $f(1-0) = 2 \neq f(1+0) = 0$ 得 $x = 1$ 为间断点, 应选(B).

(3) 【答案】 (C).

【解】 由 $AB = O$ 得 $r(A) + r(B) \leq 3$,

因为 $B \neq O$, 所以 $r(B) \geq 1$, 从而 $r(A) < 3$, 于是 $|A| = 0$,

$$\text{由 } \begin{vmatrix} \lambda & 1 & \lambda^2 \\ 1 & \lambda & 1 \\ 1 & 1 & \lambda \end{vmatrix} = 0 \text{ 得 } \lambda = 1,$$

由 $A \neq O$ 得 $r(A) \geq 1$, 从而 $r(B) \leq 2 < 3$, 于是 $|B| = 0$, 应选(C).

(4) 【答案】 (B).

【解】 $|A| = [(n-1)a + 1](1-a)^{n-1}$,

由 $r(A) < n$ 得 $a = 1$ 或 $a = \frac{1}{1-n}$,

当 $a = 1$ 时, $r(A) = 1 \neq n-1$, 故 $a = \frac{1}{1-n}$, 应选(B).

(5) 【答案】 (A).

【解】 因为 $F(x), F_1(x), F_2(x)$ 都是分布函数, 所以 $F(+\infty) = F_1(+\infty) = F_2(+\infty) = 1$,

从而 $a-b=1$, 应选(A).

$$\text{三、【解】 } \frac{\partial z}{\partial x} = 2xe^{-\arctan \frac{y}{x}} - (x^2 + y^2)e^{-\arctan \frac{y}{x}} \cdot \frac{1}{1 + \frac{y^2}{x^2}} \cdot \left(-\frac{y}{x^2}\right) = (2x+y)e^{-\arctan \frac{y}{x}},$$

$$\frac{\partial z}{\partial y} = 2ye^{-\arctan \frac{y}{x}} - (x^2 + y^2)e^{-\arctan \frac{y}{x}} \cdot \frac{1}{1 + \frac{y^2}{x^2}} \cdot \frac{1}{x} = (2y-x)e^{-\arctan \frac{y}{x}},$$

$$\text{则 } dz = e^{-\arctan \frac{y}{x}} [(2x+y)dx + (2y-x)dy].$$

$$\frac{\partial^2 z}{\partial x \partial y} = e^{-\arctan \frac{y}{x}} - (2x+y)e^{-\arctan \frac{y}{x}} \cdot \frac{1}{1 + \frac{y^2}{x^2}} \cdot \frac{1}{x} = \frac{y^2 - xy - x^2}{x^2 + y^2} e^{-\arctan \frac{y}{x}}.$$

四、【解】 方法一 令 $\begin{cases} x = r \cos \theta, \\ y = r \sin \theta \end{cases} \left(-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}, 0 \leq r \leq \cos \theta \right)$, 则

$$\iint_D \sqrt{x} dx dy = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_0^{\cos \theta} r^{\frac{3}{2}} \sqrt{\cos \theta} dr = \frac{4}{5} \int_0^{\frac{\pi}{2}} \cos^3 \theta d\theta = \frac{8}{15}.$$

方法二 $D = \{(x, y) \mid 0 \leq x \leq 1, -\sqrt{x-x^2} \leq y \leq \sqrt{x-x^2}\}$, 则

$$\begin{aligned} \iint_D \sqrt{x} \, dx \, dy &= \int_0^1 \sqrt{x} \, dx \int_{-\sqrt{x-x^2}}^{\sqrt{x-x^2}} dy = 2 \int_0^1 x \sqrt{1-x} \, dx \\ &\stackrel{\sqrt{1-x}=t}{=} 2 \int_1^0 (1-t^2) \cdot (-2t^2) dt = 4 \int_0^1 (t^2 - t^4) dt = 4 \left(\frac{1}{3} - \frac{1}{5} \right) = \frac{8}{15}. \end{aligned}$$

五、【解】 以连续复利计息, 这批酒窖藏 t 年末售出的总收入 R 的现值为

$$A(t) = R e^{-rt} = R_0 e^{\frac{2}{5}\sqrt{t}-rt},$$

$$\text{令 } \frac{dA}{dt} = R_0 e^{\frac{2}{5}\sqrt{t}-rt} \cdot \left(\frac{1}{5\sqrt{t}} - r \right) = 0, \text{ 得 } t = \frac{1}{25r^2}.$$

因为当 $0 < t < \frac{1}{25r^2}$ 时, $\frac{dA}{dt} > 0$; 当 $t > \frac{1}{25r^2}$ 时, $\frac{dA}{dt} < 0$,

所以当 $t = \frac{1}{25r^2}$ 时, 总收入的现值最大,

当 $r = 0.06$ 时, $t = \frac{100}{9} \approx 11$ (年).

六、【证明】 令 $h(x) = e^x$, $h'(x) = e^x \neq 0$ ($a < x < b$), 由柯西中值定理, 存在 $\eta \in (a, b)$, 使得

$$\frac{f(b) - f(a)}{h(b) - h(a)} = \frac{f'(\eta)}{h'(\eta)}, \text{ 即 } \frac{f(b) - f(a)}{e^b - e^a} = \frac{f'(\eta)}{e^\eta},$$

$$\text{从而 } \frac{f(b) - f(a)}{b - a} = \frac{e^b - e^a}{b - a} \cdot \frac{f'(\eta)}{e^\eta},$$

再由拉格朗日中值定理, 存在 $\xi \in (a, b)$, 使得 $f'(\xi) = \frac{f(b) - f(a)}{b - a}$,

$$\text{即 } f'(\xi) = \frac{e^b - e^a}{b - a} \cdot \frac{f'(\eta)}{e^\eta}, \text{ 故 } \frac{f'(\xi)}{f'(\eta)} = \frac{e^b - e^a}{b - a} e^{-\eta}.$$

$$\text{七、【解】 (1) 由 } \begin{cases} y = nx^2 + \frac{1}{n}, \\ y = (n+1)x^2 + \frac{1}{n+1}, \end{cases} \text{ 得 } x^2 = \frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)}, \text{ 则 } a_n = \frac{1}{\sqrt{n(n+1)}},$$

由对称性得

$$\begin{aligned} S_n &= 2 \int_0^{a_n} \left[nx^2 + \frac{1}{n} - (n+1)x^2 - \frac{1}{n+1} \right] dx = 2 \int_0^{a_n} \left[\frac{1}{n(n+1)} - x^2 \right] dx \\ &= \frac{4}{3} \frac{1}{n(n+1) \sqrt{n(n+1)}}. \end{aligned}$$

$$(2) \sum_{n=1}^{\infty} \frac{S_n}{a_n} = \frac{4}{3} \sum_{n=1}^{\infty} \frac{1}{n(n+1)},$$

$$\text{部分和 } S'_n = \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \cdots + \frac{1}{n(n+1)} = 1 - \frac{1}{n+1},$$

$$\text{因为 } \lim_{n \rightarrow \infty} S'_n = 1, \text{ 所以 } \sum_{n=1}^{\infty} \frac{S_n}{a_n} = \frac{4}{3} \sum_{n=1}^{\infty} \frac{1}{n(n+1)} = \frac{4}{3}.$$

$$\text{八、【解】 } V(t) = \pi \int_1^t f^2(x) dx,$$

$$\text{由题意得 } \frac{\pi}{3} [t^2 f(t) - f(1)] = \pi \int_1^t f^2(x) dx, \text{ 即 } t^2 f(t) - f(1) = 3 \int_1^t f^2(x) dx,$$

两边对 t 求导得

$$2tf(t) + t^2 f'(t) = 3f^2(t),$$

从而 $y = f(x)$ 所满足的微分方程为 $2xy + x^2y' = 3y^2$.

由 $2xy + x^2y' = 3y^2$ 得 $y' = -2\frac{y}{x} + 3\left(\frac{y}{x}\right)^2$,

令 $u = \frac{y}{x}$, 代入得 $u + x \frac{du}{dx} = -2u + 3u^2$, 整理得 $\frac{du}{u(u-1)} = 3\frac{dx}{x}$,

积分得 $\ln \frac{u-1}{u} = \ln x^3 + C_1$, 解得 $\frac{u-1}{u} = C_2 x^3$, 通解为 $y - x = C_2 x^3 y$,

将初始条件代入得 $C_2 = -1$, 所求的解为 $y - x = -x^3 y$.

九、【解】 (1) 由 $\alpha^T \beta = 0$ 得 $\alpha \perp \beta$,

则 $A^2 = \alpha \beta^T \cdot \alpha \beta^T = \alpha (\beta^T \alpha) \beta^T = O$.

(2) 令 $AX = \lambda X$ (X 为非零向量),

由 $A^2 X = \lambda^2 X = O$ 得 $\lambda_1 = \lambda_2 = \cdots = \lambda_n = 0$.

不妨设 $a_1 \neq 0, b_1 \neq 0$, 矩阵 A 的属于特征值 $\lambda_1 = \lambda_2 = \cdots = \lambda_n = 0$ 的特征向量即为方程组 $(0E - A)X = O$ 的非零解,

由 $0E - A \rightarrow \begin{pmatrix} 1 & \frac{b_2}{b_1} & \cdots & \frac{b_n}{b_1} \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}$ 得 $\lambda = 0$ 对应的线性无关的特征向量为

$$\alpha_1 = \left(-\frac{b_2}{b_1}, 1, 0, \cdots, 0\right)^T, \alpha_2 = \left(-\frac{b_3}{b_1}, 0, 1, \cdots, 0\right)^T, \cdots, \alpha_{n-1} = \left(-\frac{b_n}{b_1}, 0, \cdots, 0, 1\right)^T,$$

故属于 $\lambda = 0$ 的全部特征向量为 $k_1 \alpha_1 + k_2 \alpha_2 + \cdots + k_{n-1} \alpha_{n-1}$ ($k_1, k_2, \cdots, k_{n-1}$ 不全为零).

十、【解】 由 $|\lambda E - A| = \begin{vmatrix} \lambda - 1 & 0 & -1 \\ 0 & \lambda - 2 & 0 \\ -1 & 0 & \lambda - 1 \end{vmatrix} = \lambda(\lambda - 2)^2$, 得矩阵 A 的特征值为 $\lambda_1 = 0, \lambda_2 = \lambda_3 = 2$,

则存在正交矩阵 Q , 使得 $Q^T A Q = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$,

从而 $Q^T (kE + A) Q = \begin{pmatrix} k & 0 & 0 \\ 0 & k+2 & 0 \\ 0 & 0 & k+2 \end{pmatrix}$.

于是 $Q^T B Q = Q^T (kE + A)^2 Q = Q^T (kE + A) Q \cdot Q^T (kE + A) Q$

$$= \begin{pmatrix} k^2 & 0 & 0 \\ 0 & (k+2)^2 & 0 \\ 0 & 0 & (k+2)^2 \end{pmatrix},$$

得

$$A = \begin{pmatrix} k^2 & 0 & 0 \\ 0 & (k+2)^2 & 0 \\ 0 & 0 & (k+2)^2 \end{pmatrix}.$$

因为矩阵 A 的特征值为 $\lambda_1 = 0, \lambda_2 = \lambda_3 = 2$,

所以 $B = (kE + A)^2$ 的特征值为 $\lambda_1 = k^2, \lambda_2 = \lambda_3 = (k+2)^2$.

显然 B 为实对称矩阵, 故 B 为正定矩阵的充分必要条件是 $k^2 > 0, (k+2)^2 > 0$, 即 $k \neq 0, k \neq -2$.

十一、【解】 设获利为 Z , 根据题意得

$$Z = \begin{cases} 1\,000Y, & Y \leq X, \\ 1\,000X + 500(Y - X), & Y > X, \end{cases}$$

因为 X, Y 相互独立且都服从 $[10, 20]$ 上的均匀分布, 所以 X, Y 的联合概率密度为

$$f(x, y) = \begin{cases} \frac{1}{100}, & 10 \leq x \leq 20, 10 \leq y \leq 20, \\ 0, & \text{其他}, \end{cases}$$

令 $D_1 = \{(x, y) \mid 10 \leq x \leq 20, 10 \leq y \leq x\}$, $D_2 = \{(x, y) \mid 10 \leq x \leq 20, x \leq y \leq 20\}$,

$$\begin{aligned} \text{则 } E(Z) &= \frac{1}{100} \iint_{D_1} 1\,000y \, dx \, dy + \frac{1}{100} \iint_{D_2} 500(x + y) \, dx \, dy \\ &= 10 \int_{10}^{20} dx \int_{10}^x y \, dy + 5 \int_{10}^{20} dx \int_x^{20} (x + y) \, dy \\ &= 5 \int_{10}^{20} (x^2 - 100) \, dx + 5 \int_{10}^{20} \left[x(20 - x) + \frac{400 - x^2}{2} \right] \, dx \\ &= 14\,166.67 (\text{元}). \end{aligned}$$

十二、【解】(1) 令 $A_i = \{\text{取第 } i \text{ 个地区}\} (i = 1, 2, 3)$,

$B_i = \{\text{第 } i \text{ 次取到女生报名表}\} (i = 1, 2)$,

显然 $P(A_1) = P(A_2) = P(A_3) = \frac{1}{3}$,

$$P(B_1 | A_1) = \frac{3}{10}, \quad P(B_1 | A_2) = \frac{7}{15}, \quad P(B_1 | A_3) = \frac{5}{25},$$

$$\text{则 } p = P(B_1) = \sum_{i=1}^3 P(A_i)P(B_1 | A_i) = \frac{1}{3} \left(\frac{3}{10} + \frac{7}{15} + \frac{5}{25} \right) = \frac{29}{90}.$$

$$(2) q = P(B_1 | \overline{B_2}) = \frac{P(B_1 \overline{B_2})}{P(\overline{B_2})},$$

$$\text{由 } P(B_2) = P(B_1) = \frac{29}{90}, \text{ 得 } P(\overline{B_2}) = 1 - P(B_2) = \frac{61}{90},$$

$$\text{由 } P(B_1 \overline{B_2} | A_1) = \frac{3}{10} \times \frac{7}{9} = \frac{7}{30},$$

$$P(B_1 \overline{B_2} | A_2) = \frac{7}{15} \times \frac{8}{14} = \frac{8}{30},$$

$$P(B_1 \overline{B_2} | A_3) = \frac{5}{25} \times \frac{20}{24} = \frac{1}{5} \times \frac{5}{6} = \frac{5}{30},$$

$$\begin{aligned} \text{得 } P(B_1 \overline{B_2}) &= P(A_1)P(B_1 \overline{B_2} | A_1) + P(A_2)P(B_1 \overline{B_2} | A_2) + P(A_3)P(B_1 \overline{B_2} | A_3) \\ &= \frac{1}{3} \left(\frac{7}{30} + \frac{8}{30} + \frac{5}{30} \right) = \frac{2}{9}, \end{aligned}$$

$$\text{故 } q = \frac{2}{9} \times \frac{90}{61} = \frac{20}{61}.$$